

AssessClear AI — Pre-Pilot Analysis & Strategic Evaluation

Context Learning, Pre-Pilot Cross-District Findings, Root-Cause Diagnostics & Pilot Readiness Framework

ROLE / AUTHOR	SCOPE & GEOGRAPHY	EVALUATED SAMPLE	TARGET LAUNCH
Senior Product Evaluation Specialist	9 Schools (Katni, Indore, Bhopal, Raisen)	~351 Student Sheets · Grades 6–8	Madhya Pradesh Pilot (2026)

1. Executive Summary & Readiness Verdict

This evaluation report provides an independent, evidence-based assessment of the **AssessClear** diagnostic assessment platform and its **Sakhi AI** WhatsApp companion, analyzing primary field reports from pre-pilot deployments across Indore, Bhopal, Raisen, and Katni districts in Madhya Pradesh.

Pre-Pilot Readiness Verdict: REQUIRES ATTENTION BEFORE PILOT (Conditional Green Light)

While the core pedagogical value of multi-tiered diagnostic feedback (1.0, 0.75, 0.40, 0.0) is validated, the system exhibits critical functional defects (question segmentation failure, digit misrecognition, rigid exact-matching) and operational bottlenecks ("double effort" workflow taking 80–90 min per 50 sheets) that must be resolved prior to full pilot deployment.

Key Strengths Validated in Field

- Multi-Tiered Diagnostic Feedback:**
Moves beyond binary scoring to actionable remediation (*What was done well, Think about this, Next step*).
- Partial Attempt Recognition:**
Successfully rewards constructive mathematical attempts (0.40 / 0.75 weights).
- State Hindi Curriculum Alignment:**
Direct ingestion of handwritten Hindi answers fits MP government school needs.

Critical Failure Points Detected

- Question Skipping (Q1/Q2):**
AI systematically dropped top/bottom questions on clean pages (evaluated only 7 of 10 questions).
- High Teacher Overhead:**
1–3 min review time per child turns time-saving promise into an administrative burden.
- Offline Vulnerability:**
Rural schools (e.g. MS Kharbai) stalled completely due to lack of offline caching.

2. AssessClear Architectural Context & Knowledge Model

AssessClear addresses the diagnostic deficit in government classrooms by transforming a mobile photo of a handwritten student worksheet into an automated diagnostic report in seconds at a target compute cost of **₹0.12 per evaluation** (utilizing multimodal `gemini-3.1-flash-lite` on Google Cloud Run Mumbai).

System Dimension	Designed / Intended Architecture	Pre-Pilot Field Performance Reality
Core Vision Pipeline	Client-side auto-crop & compression + Gemini multimodal OCR.	Vision model skips Q1/Q2; digit misrecognition (9996 read as 3226); MCQ option D read as B .
Scoring Logic	4-tier weighted rubric (1.0, 0.75, 0.40, 0.0) + distractor mapping.	Penalizes valid semantic synonyms (e.g., marks 3 months incorrect when key specifies 90 days).
Operational Ingestion	Rapid Batch Upload (up to 10 sheets) or One-by-One capture.	Rapid batch works well for single pages; One-by-One capture has page omission issues.
Network Resilience	Mobile-responsive PWA optimized for 3G/4G connectivity.	Zero offline resilience; total freeze in zero-connectivity schools (MS Kharbai, Raisen).
Dashboard & Storage	Live SLO heatmaps, student cohort growth & cloud storage.	Dashboard counts questions instead of copies; hard cap of ~200 items drops historical evaluations.
Remedial Companion	Sakhi AI WhatsApp Companion (+91 8796104226) for 7 resources.	Promising concept, but requires automated linkage to AssessClear diagnostic color bands.

3. Cross-District Evidence Synthesis (9 Schools, ~351 Students)

Cross-report analysis reveals consistent patterns across all participating districts:

District / Region	Coverage & Scale	Key Field Observations	Critical Defects Logged
Indore (Nandbag & Subhash Nagar)	2 Schools 100 Students (60 Math, 40 Hindi) DIET Observer	• Rapid upload effective for ≤10 sheets. • 20–30s AI processing time. • DIET Observer noted tool is an assistant, not a replacement.	• Top/bottom questions (Q1, Q2) skipped (10 attempted → 7 evaluated). • Rigid logic rejected "3 months" for "90 days". • Incorrect arithmetic box sum marked as correct. • Dashboard hid Hindi and Grades 7–8 data.
Bhopal & Raizen (5 Schools incl. MS Kharbai)	5 Schools ~160 Students Grades 6–8 (Math & Hindi)	• Structured 3-part feedback easy to grasp. • Scanning directly inside app saved device photo storage. • Hindi OCR validated as relevant.	• Zero connectivity in MS Kharbai blocked app usage completely. • Reviewing 30+ students individual feedback took >80 min. • Downloading individual PDFs fills phone storage. • Feedback focused on "what" rather than "how to improve".
Katni (EPES Purwar & MS Devri Hatai)	2 Schools 91 Students (32 Purwar, 59 Hatai) Grade 6 Math/Hindi	• Step-by-step remedial handholding recognized as high potential. • Partial attempt recognition encouraged student effort.	• Dashboard showed 29 evaluations for 3 copies. • Session codes hard to retrieve once navigating away. • Digit misrecognition (9996 → 3226). • Corrected pencil answers ignored. • 200-evaluation cap purged older test history.

4. User Experience & Workflow Friction Analysis

An evaluation of the complete user journey (**Entry** → **Setup** → **Capture** → **AI Evaluation** → **Review** → **Classroom Action**) identifies the following primary friction points:

The "Double Effort" Dilemma (Teacher Feedback):

Teachers reported that instead of saving time, using AssessClear in its current form introduces 4 extra manual steps:

1. Photographing each student worksheet individually or in small batches.
2. Waiting for cloud processing (20–30 sec per sheet).
3. Reviewing multi-line diagnostic feedback on screen (1–3 min per sheet).
4. Manually transcribing the AI remarks onto the physical paper for the student.

Total Time for 40 Students = 90–120 minutes. Without an aggregated Class-Level Summary Dashboard, routine classroom adoption is unsustainable.

5. Root-Cause Analysis for Critical Failures

Observed Defect	Primary Root Cause	Severity	Technical & Pedagogical Remedy
Q1 / Q2 Skipping	Client-side auto-crop applies fixed bounding-box margins that clip top and bottom question headers.	CRITICAL	Expand crop boundary margins and add a backend question-count assertion rule against paper schema.
Rigid Semantic Matching ("3 mos" vs "90 days")	Gemini system prompt lacked explicit pedagogical instructions for unit conversion & synonym equivalence.	HIGH	Update LLM evaluation prompt with semantic tolerance guidelines and conceptual equivalence rules.
Evaluation Count Mismatch (29 for 3 copies)	Firestore schema tracks individual question assessment sub-documents rather than student copy sessions.	HIGH	Refactor schema to aggregate by <code>student_submission_id</code> and paginate queries past 200 items.
Blurry Image Hallucination	Missing pre-flight image quality validation before sending payload to the multimodal API.	CRITICAL	Implement client-side Laplacian blur/sharpness check; prompt user to retake photo if score < threshold.
Network Lockout in Rural Areas	PWA lacks Service Worker caching and local IndexedDB queueing.	CRITICAL	Implement offline-first storage to cache scanned sheets and sync in background when online.

6. Master Strategic Prioritization Matrix (P0 / P1 / P2 / P3)

Pri.	Issue & Description	Field Evidence	Impact	Sev.	Effort	Mandatory Action
P0	Question Segmentation Fix	Q1 & Q2 skipped in Indore & Katni (10 → 7)	Corrupts diagnostic score and attempt count	CRIT.	Med	Adjust crop padding & enforce question schema count assertions.
P0	Offline Ingestion & Sync	Complete failure in MS Kharbai (Raisen)	Blocks field deployment in rural schools	CRIT.	High	Implement Service Worker & IndexedDB local submission queue.
P0	Image Quality Gatekeeper	AI hallucinating on blurry sheets (Katni)	False diagnostic data entered into system	CRIT.	Low	Add client-side blur detection; prompt immediate re-take.
P1	Class-Level Summary View	80–90 min review time across all districts	Teacher burnout & pre-pilot rejection	HIGH	Med	Build one-screen class misconception heatmap & priority list.
P1	Teacher Score Override	DIET Observer note & Katni/Indore reports	Teacher unable to fix incorrect AI verdicts	HIGH	Low	Add 1-click manual score/verdict correction button on UI.
P1	Session History Management	Teachers losing session codes in Katni/Bhopal	Orphaned evaluations & duplicate sessions	HIGH	Low	Provide active/recent sessions list on user dashboard.
P1	Dashboard Metric Schema Fix	29 evals for 3 copies; 200 item cap (Katni)	Loss of historical data & reporting errors	HIGH	Med	Redefine metric to student copies; add query pagination.
P2	Actionable Feedback Prompts	Generic "practice antonyms" advice (Raisen)	Low classroom utility for teachers	MED.	Low	Update LLM prompt to include specific pedagogical drills.
P2	Hindi PDF Export Alignment	Student feedback omitted from exported PDF	Inability to share feedback with students	MED.	Low	Ensure exported PDF includes both Teacher & Student views.

7. Proposed Pilot Monitoring Framework & Research Questions

Key Quantitative KPI Benchmarks

- **Question Detection Fidelity:** > 99.0% of printed questions detected and scored.
- **AI Scoring Accuracy:** > 95.0% alignment between AI grade and expert teacher review.
- **Teacher In-App Review Overhead:** < 15 seconds per student via Class Summary view.
- **Offline Sync Integrity:** 100% of locally cached worksheets synced upon reconnection.
- **Session Completion Rate:** > 90% of initiated classroom assessment batches completed.

Core Pilot Research Questions

1. Does automated diagnostic grading directly accelerate spot remediation compared to legacy exams?
2. Can the Class-Level Summary UI reduce teacher workload sufficiently to achieve >75% weekly adoption?
3. How effectively does Sakhi AI WhatsApp companion close diagnostic learning gaps identified by AssessClear?
4. What is the real-world serverless compute cost at district scale (10,000+ evals/day)?

8. Final Analytical Conclusion

AssessClear represents a **transformative diagnostic innovation** for government school education, replacing delayed, high-overhead manual grading with real-time, multi-tiered misconception analysis. The pre-pilot exercise in Madhya Pradesh successfully proved the viability of mobile multimodal grading at ₹0.12/eval.

By executing the **P0 blocker resolutions** (crop boundary correction, image quality gate, offline sync, and class-level summary screens) before full rollout, project leadership can ensure a high-fidelity, teacher-friendly, and statistically robust pilot deployment across the state.